

SDS 315 Homework 8

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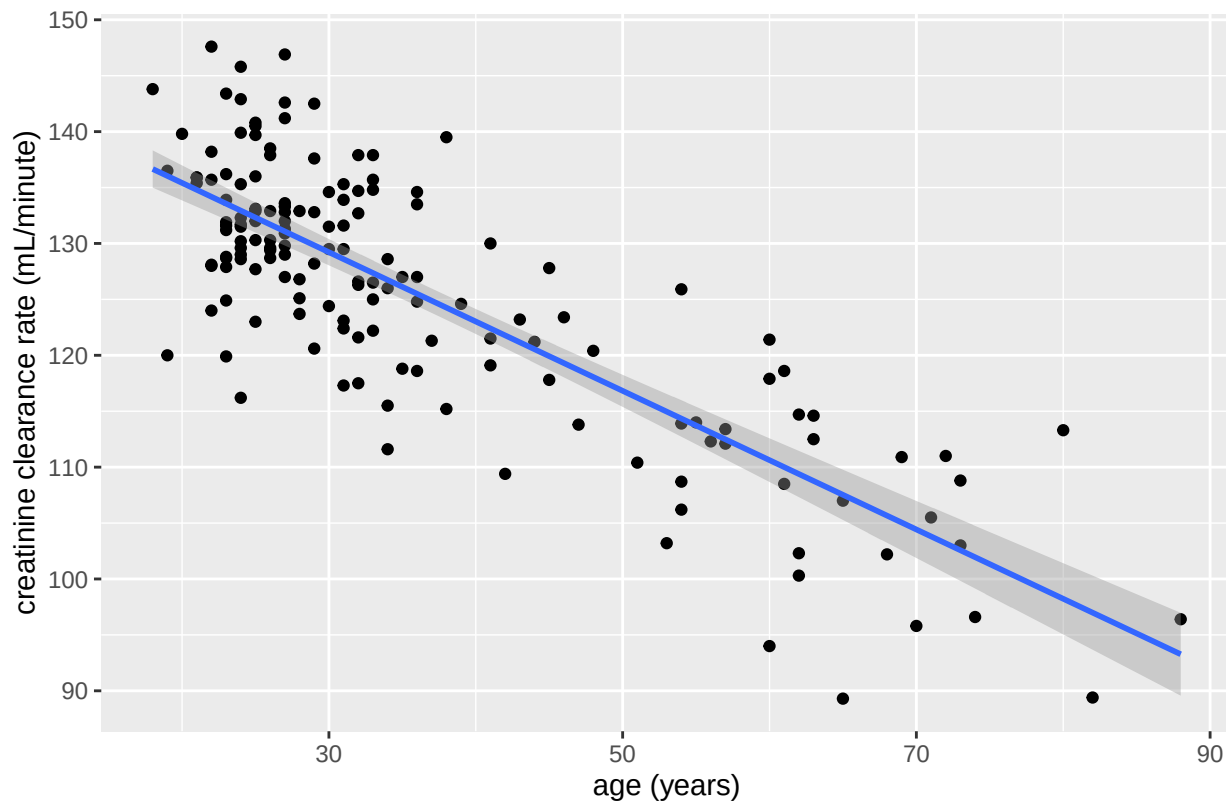
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April 14th, 2025

Problem 1: regression warm up

Creatinine clearance rate change with age



```
## (Intercept)      age
## 147.8129158 -0.6198159
```

A) What creatinine clearance rate should we expect for a 55-year-old?

A 55-year-old is expected to have a creatinine clearance rate of 113.7 mL/minute based on the linear regression model.

This was calculated using the regression equation derived from the model:

$$\text{creatinineClearanceRate} = 147.8 - 0.62 * \text{age}$$

expected rate of 55 year old = $147.8 - 0.62 * 55 = 113.7$ mL/min

B) How does creatinine clearance rate change with age?

The creatinine clearance rate change with age is represented by the slope of the regression line, which is -0.62 mL/minute per year. This means that for every additional year of age, the creatinine clearance rate will decrease on average by 0.62 mL/minute.

C) Whose creatinine clearance rate is healthier (higher) for their age: a 40-year-old with a rate of 135, or a 60-year-old with a rate of 112?

A 40-year-old with a clearance rate of 135

A 60-year-old with a clearance rate of 112

Using the regression equation:

$$\text{creatinineClearanceRate} = 147.8 - 0.62 * \text{age}$$

Calculate the expected clearance rate for each and compare the residuals (actual - expected):

Calculate expected values: Expected for age 40:

$$147.8 - 0.62 \times 40 = 123$$

Residual:

$$135 - 123 = 12$$

Expected for age 60:

$$147.8 - 0.62 \times 60 = 110.6$$

Residual:

$$112 - 110.6 = 1.4$$

Since higher creatinine clearance rates indicate better kidney function, the 40-year-old has a healthier creatinine clearance rate for their age. Their clearance rate is 12 mL/min above the expected value, whereas the 60-year-old's clearance rate is only 1.4 mL/min above their expected value. Therefore, the 40-year-old exceeds their expected rate by a larger margin, indicating better kidney function relative to their age.

Problem 2: Modeling disease growth

1. Estimated growth rate and doubling time for Italy, with 95% bootstrapped confidence intervals for each

Italy

##	name	lower	upper level	method	estimate
##	2 days_since_first_death	0.1595504	0.2073149	0.95 percentile	0.183218

By using a linear model I was able to understand how daily COVID-19 deaths increased in Italy over time. The estimated growth rate for Italy was about 0.183, indicating that the deaths increased by 18.3 % every day. The estimated doubling time was 3.8 days. The 95% bootstrapped confidence intervals for the growth rate ranged from around 0.159 to 0.208. The 95% bootstrapped confidence intervals for the doubling time ranged from around 4.4 days and 3.4 days.

2. Estimated growth rate and doubling time for Spain, with 95% bootstrapped confidence intervals for each

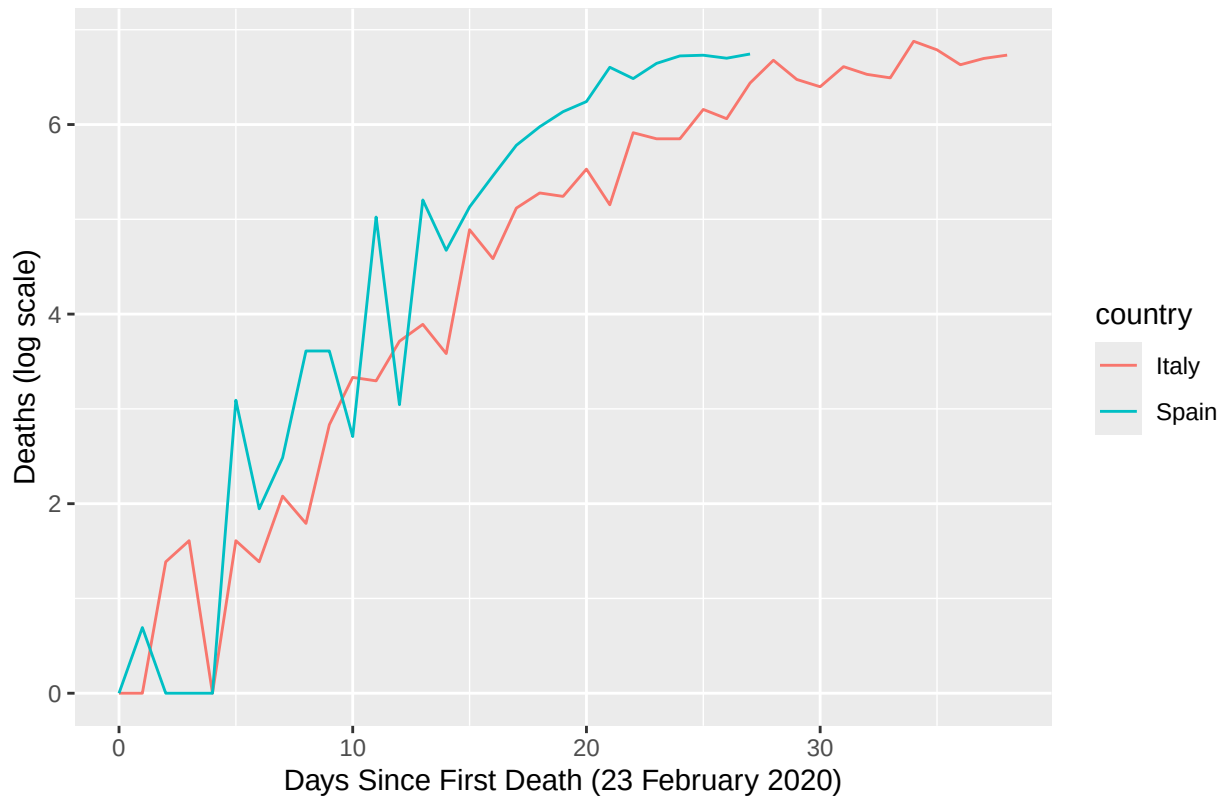
Spain

##	name	lower	upper level	method	estimate
##	2 days_since_first_death	0.2350098	0.3164495	0.95 percentile	0.2762447

By using a linear model I was able to understand how daily COVID-19 deaths increased in Spain over time. The estimated growth rate for Spain was 0.276, indicating that the deaths increased by 27.6% every day. The estimated doubling time was 2.5 days. The 95% bootstrapped confidence intervals for the growth rate ranged from around 0.235 and 0.318. The 95% bootstrapped confidence intervals for the doubling time ranged from around 3.0 days and 2.2 days.

3.

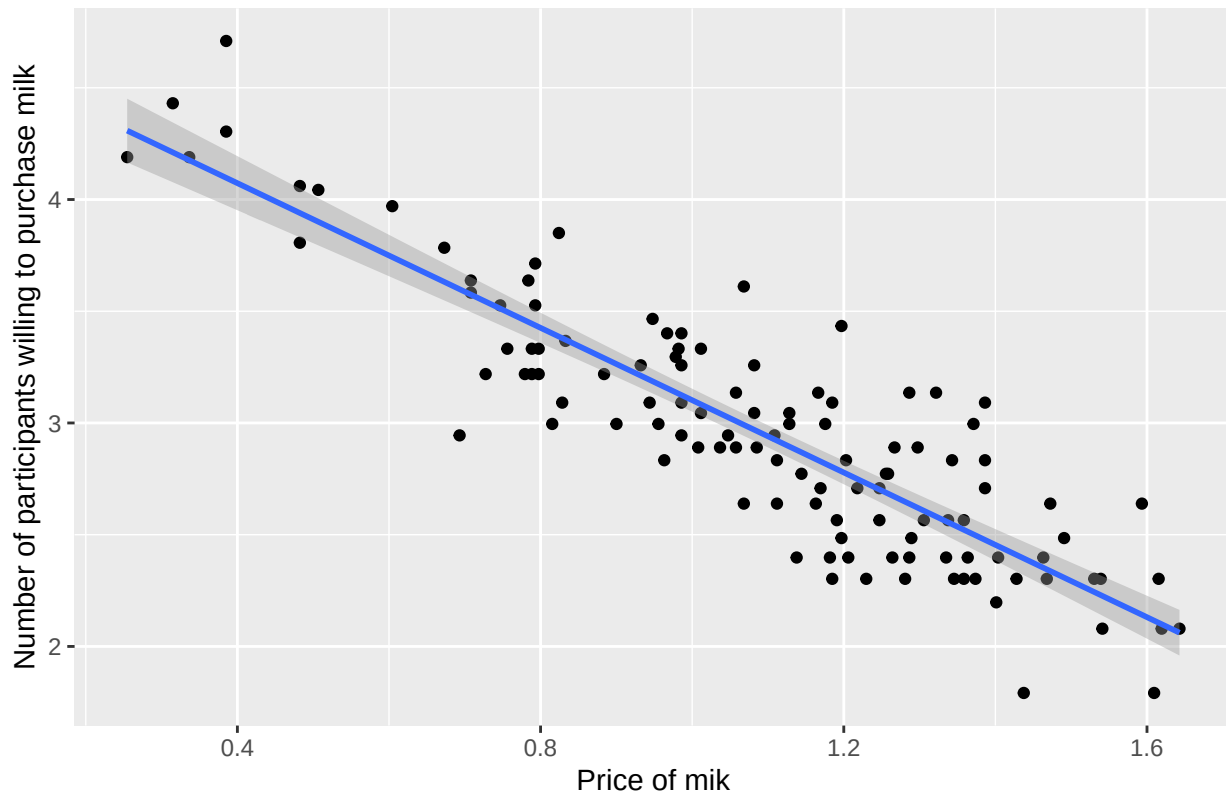
Daily COVID-19 Deaths Over Time Fit on log scale



The graph represents the daily COVID deaths that happened over time starting from February 23rd, 2020 in Italy and Spain. The graph provides a visual comparison of the rate of increasing deaths between Italy and Spain. Spain has a more steep slope than Italy indicating that Spain had a higher growth rate than Italy. This is supported by Spain's greater growth rate and shorter doubling time as calculated earlier.

Problem 3: price elasticity of demand

Price elasticity of demand for milk



```
##      name      lower  upper level    method estimate
## 2 log_price -1.773052 -1.45163  0.95 percentile -1.618578
```

The estimated price elasticity of demand for milk is -1.62, with a 95% bootstrapped confidence interval ranging from around -1.77 to -1.46.

To estimate this value, I used a power-law model of the form:

$$Q = K \cdot P^\beta$$

By taking the logarithm of both sides, I transformed it into a linear model:

$$\log(Q) = \log(K) + \beta \cdot \log(P)$$

I fit this transformed model using linear regression of $\log(\text{sales})$ on $\log(\text{price})$ and bootstrapped the model 10,000 times to obtain a confidence interval for the elasticity estimate β . Since the slope of this regression corresponds to the price elasticity, a value of -1.62 implies that a 1% increase in price leads to a 1.62% decrease in milk sales, indicating that demand is elastic. Additionally, because the confidence interval does not include 0, the result is statistically significant and confirms that price of milk has a meaningful impact on its demand.